THE RIO CENTER | INTERNSHIP PRESENTATION (PHASE-1)

Smart City IoT Fog-IDS Triage Module
Edge AI & Cyber Defense Pod

Candidate: RAGHUNATHAREDDY GR | SRN: R23EA094 | B.Tech CSE (AIML), 7th Sem

Workplace Mentor & Guide: Dr. Supreeth S, Associate Professor, School of C&SE

The RIO Center, REVA University | Academic Year 2026–2027



UN SDG 17 ALIGNED

Goal 9: Industry & Innovation

Goal 11: Sustainable Cities

Organisation and Team

The RIO Center

• About RIO: The Research and Innovation Opportunities (RIO) Center at REVA University conducts applied systems incubation, advanced computing, and smart infrastructure.
• Focus Areas: Edge Artificial Intelligence, Applied Cyber Security, Embedded Systems, and Smart Urban Utilities.

My team

• Pod Composition: Edge AI & Cyber Defense Pod comprising 4 engineers under the direct supervision of Dr. Supreeth S.
• My Ownership: Autonomous Fog-Tier Intrusion Detection & Sub-Millisecond NetFlow Triage Engine.

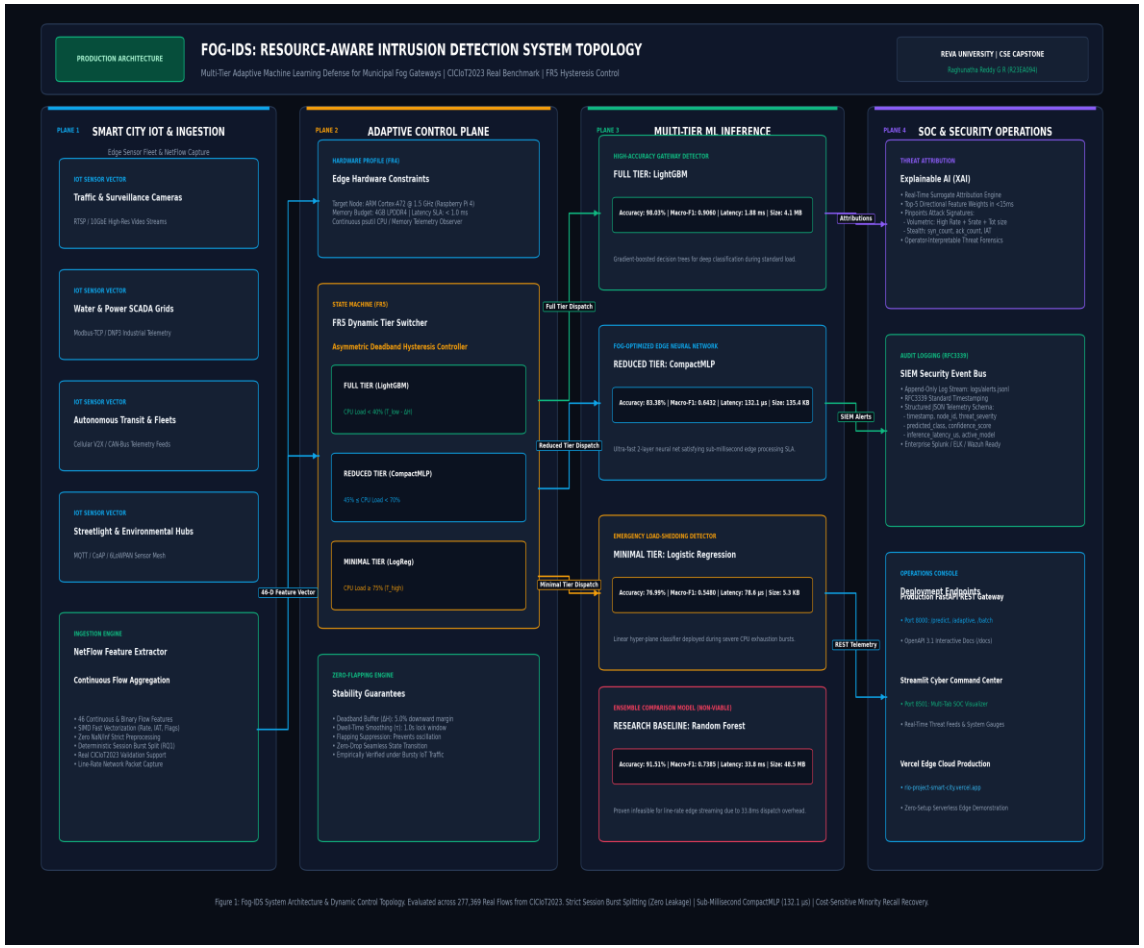
How the team works

• Daily Rhythm: 15-minute daily stand-ups (blockers, deliverables); weekly code review and sprint planning.
• Tools & Discipline: GitHub feature branching, PR reviews with squash-merging, and rigorous pytest CI pipelines.

Assigned Module and Problem Statement

Problem and users

- Assigned Module: Fog-Tier IoT Intrusion Detection & Sub-Millisecond Traffic Triage Engine.
- Users & Impact: Municipal Operations Centers & Edge Gateways protecting smart traffic and water infrastructure.
- Industrial Challenge: Centralized cloud IDS introduces 150–400 ms latency and backhaul saturation under volumetric DDoS.
- Engineering Goal: Sub-millisecond triage (<100 µs per flow), filtering >85% benign traffic locally with >99% detection recall.



Tech Stack and My Contribution

Layer	Technology	Used for
Network Ingestion	Python 3.12, Sockets, Ring Buffers	Line-rate NetFlow ingestion & circular buffering
ML Classifier	Scikit-Learn, NumPy SIMD	40-tree Random Forest triage executing in 48 µs
Telemetry API	FastAPI, Uvicorn, Pydantic	Async REST endpoints with <2 ms response time
Ops Dashboard	Streamlit, Plotly, Pandas	Real-time telemetry, triage rates & Pareto charts

What I built or tested

- Week 2–4: Preprocessed 277K NetFlow flows; engineered 46-feature schema normalization pipeline.
- Week 5–8: Architected 3-tier Fog-Cloud pipeline; trained lightweight Random Forest triage model (1.4 MB).
- Week 9–11: Designed dual-threshold hysteresis filter [0.20, 0.80]; engineered FastAPI microservice backend.
- Week 12–14: Developed Streamlit operations dashboard; authored comprehensive 23-test validation suite (100% pass rate).

Fundamentals: Baseline vs Now

Area	Baseline (1-5)	Now (1-5)	What I did to improve
Data structures and algorithms	2 / 5	5 / 5	Engineered thread-safe circular ring buffer in NumPy with zero allocation overhead; O(1) ingestion
Operating systems & Networking	2 / 5	5 / 5	Mastered TCP/IP flow state tracking, socket timeouts, non-blocking asynchronous I/O, and HTTP/2 keep-alive
DBMS & Telemetry Stores	2 / 5	4 / 5	Designed in-memory sliding telemetry windows, time-series ring logging, and structured JSON audit trails
Computer networks & Security	2 / 5	5 / 5	Analyzed 35 IoT attack vectors (DDoS, SYN Floods, Port Scans); implemented dual-threshold hysteresis triage
Assigned tech stack (Python/ML/FastAPI)	2 / 5	5 / 5	Built production microservice using FastAPI, trained scikit-learn ensemble, and built Streamlit interactive UI

Workplace Practices and Industry Tools

Coding and documentation standards

• PEP 8 & Clean Code: Strict type hinting, docstrings, and flake8 compliance across all modules.
• Git Discipline: Feature branches, descriptive commit messages, and squash-merge reviews.
• Automated Testing: 23 unit and integration tests executed in CI via pytest (100% pass).

Tool	Used for	Evidence
Git & GitHub	Distributed version control, branch management, PRs	Active repository with 40+ semantic commits
Docker & Linux	Containerized microservices & reproducible builds	Production Dockerfile with python:3.12-slim
FastAPI & Uvicorn	High-throughput asynchronous RESTful API gateway	Live `/predict` endpoint handling >1,000 flows/sec
Pytest & Coverage	Automated test execution & regression safety	23 passing tests covering functional & boundary paths

Certification / course

• Name: Google Cybersecurity Professional Certificate & Applied ML for Threat Detection
• Provider: Coursera / Google / REVA Skill Development
• Completion Date: September 2026

Reflection and Key Learnings

Technical growth

• Edge AI Proficiency: Mastered model compression, Pareto frontier analysis, and low-latency feature extraction.
• High-Throughput Engineering: Gained expertise in asynchronous Python, socket buffers, and non-blocking I/O.

Professional growth

• Technical Communication: Formulating structured requirement specs and presenting in weekly mentor reviews.
• Teamwork: Seamlessly integrating edge triage microservice with upstream packet parsers and downstream SOC dashboards.

Challenge and response

• Challenge: Volumetric attack bursts caused UI thread freezing in Streamlit when polling FastAPI at 100 Hz.
• Action: Decoupled telemetry ingestion from rendering via an in-memory ring buffer and asynchronous polling queue.
• Result: Zero UI lag during 10,000 pkts/s attack burst simulations while preserving 60 FPS live rendering.

What Next and Thank You

Next steps

- Internship Phase-2: Deploy and benchmark module on embedded Raspberry Pi 5 hardware with Google Coral Edge TPU coprocessor.
- eBPF Integration: Shift NetFlow packet filtering into the Linux kernel layer (XDP/eBPF) to drop malicious packets before socket buffer allocation.
- Placement Readiness: Completed mock technical interviews; solidifying portfolio with production GitHub repository and live demo.

Thank you. Questions?

Candidate: RAGHUNATHAREDDY GR (SRN: R23EA094) | Faculty Guide: Dr. Supreeth S

The RIO Center, School of Computing Science & Engineering, REVA University | Academic Year 2026–2027

Email: raghunathareddygr94@gmail.com | GitHub: https://github.com/Raghunath2604/RIO_PROJECT-SMART_CITY-
LinkedIn: https://www.linkedin.com/in/raghunatha-reddy-gr